REST implementation details

ONTAP Automation

NetApp
February 11, 2024

This PDF was generated from https://docs.netapp.com/us-en/ontap-automation/rest/rest_web_services_foundation.html on February 11, 2024. Always check docs.netapp.com for the latest.

Table of Contents

REST implementation details . . . 1
- REST web services foundation . . . 1
- Basic operational characteristics . . . 3
- Input variables controlling an API request . . . 4
- Interpreting an API response . . . 9
- Asynchronous processing using the Job object . . . 11
- Object references and access . . . 12
- Performance metrics for storage resources . . . 14

REST implementation details

REST web services foundation

Representational State Transfer (REST) is a style for creating distributed web applications. When applied to the design of a web services API, it establishes a set of technologies for exposing server-based resources and managing their states. It uses mainstream protocols and standards to provide a flexible foundation for administering ONTAP clusters.

> While REST establishes a common set of technologies and best practices, the details of each API can vary based on the choices made during development. You should be aware of the design characteristics of the ONTAP REST API before using it with a live deployment.

Resources and state representation

Resources are the basic components of a web-based system. When creating a REST web services application, early design tasks include:

- Identification of system or server-based resources

  Every system uses and maintains resources. A resource can be a file, business transaction, process, or administrative entity. One of the first tasks in designing an application based on REST web services is to identify the resources.

- Definition of resource states and associated state operations

  Resources are always in one of a finite number of states. The states, as well as the associated operations used to affect the state changes, must be clearly defined.

URI endpoints

Every REST resource must be defined and made available using a well-defined addressing scheme. The endpoints where the resources are located and identified use a Uniform Resource Identifier (URI). The URI provides a general framework for creating a unique name for each resource in the network. The Uniform Resource Locator (URL) is a type of URI used with web services to identify and access resources. Resources are typically exposed in a hierarchical structure similar to a file directory.

HTTP messages

Hypertext Transfer Protocol (HTTP) is the protocol used by the web services client and server to exchange request and response messages about the resources. As part of designing a web services application, HTTP methods are mapped to the resources and corresponding state management actions. HTTP is stateless. Therefore, to associate a set of related requests and responses as part of one transaction, additional information must be included in the HTTP headers carried with the request and response data flows.

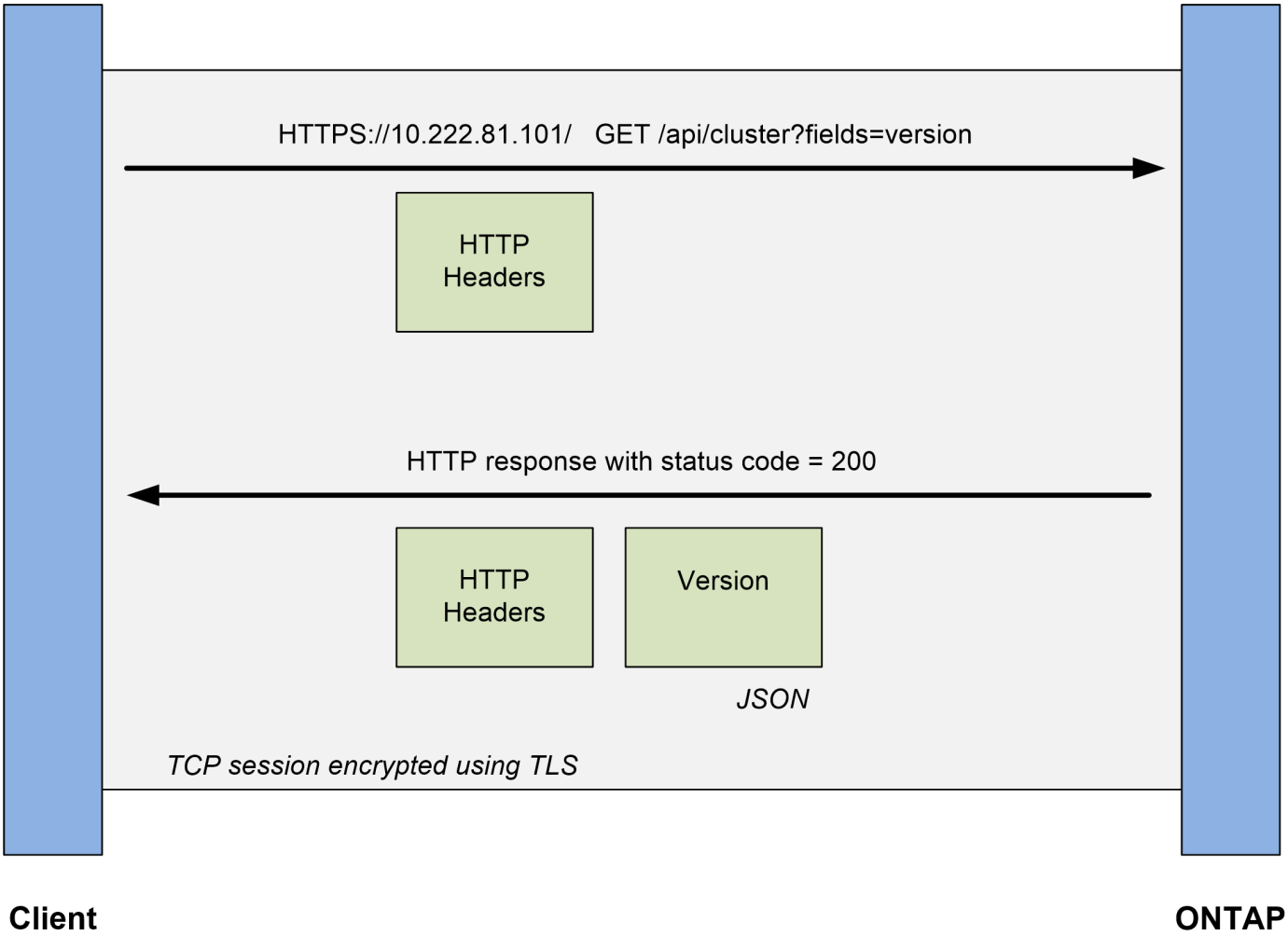
JSON formatting

Although information can be structured and transferred between a web services client and server in several ways, the most popular option is JavaScript Object Notation (JSON). JSON is an industry standard for

representing simple data structures in plain text and is used to transfer state information describing the resources. The ONTAP REST API uses JSON to format the data carried in the body of each HTTP request and response.

Typical REST API transaction

Every API transaction consists of an HTTP request and the associated response. This illustration shows how to retrieve the version of the ONTAP software used by the cluster.



HTTP request

The request sent from the client to the server consists of the following:

- GET verb
- URL path for the cluster
- Query parameter (fields)
- Request headers, including authorization

HTTP response

The response sent from the server to the client consists of the following:

- Status code 200
- Response headers

- Response body containing the cluster software version

Basic operational characteristics

While REST establishes a common set of technologies and best practices, the details of each API can vary based on the design choices.

Request and response API transaction

Every REST API call is performed as an HTTP request to the ONTAP system which generates an associated response to the client. This request/response pair is considered an API transaction. Before using the API, you should be familiar with the input variables available to control a request and the contents of the response output.

Support for CRUD operations

Each of the resources available through the ONTAP REST API is accessed based on the CRUD model:

- Create
- Read
- Update
- Delete

For some of the resources, only a subset of the operations is supported. You should review the ONTAP API documentation page at your ONTAP cluster for more information about each resource.

Object identifiers

Each resource instance or object is assigned a unique identifier when it is created. In most cases, the identifier is a 128-bit UUID. These identifiers are globally unique within a specific ONTAP cluster. After issuing an API call that creates a new object instance, a URL with the associated id value is returned to the caller in the location header of the HTTP response. You can extract the identifier and use it on subsequent calls when referring to the resource instance.

> The content and internal structure of the object identifiers can change at any time. You should only use the identifiers on the applicable API calls as needed when referring to the associated objects.

Object instances and collections

Depending on the resource path and HTTP method, an API call can apply to a specific object instance or a collection of objects.

Synchronous and asynchronous operations

There are two ways that ONTAP performs an HTTP request received from a client.

Synchronous processing

ONTAP performs the request immediately and responds with an HTTP status code of 200 or 201 if it is successful.

Every request using the methods GET, HEAD, and OPTIONS is always performed synchronously. In addition, requests that use POST, PATCH, and DELETE are designed to run synchronously if they are expected to complete in less than two seconds.

Asynchronous processing

If an asynchronous request is valid, ONTAP creates a background task to process the request and a job object to anchor the task. The 202 HTTP status is returned to the caller along with the job object. To determine final success or failure, you must retrieve the state of the job.

Requests that use the methods POST, PATCH, and DELETE are designed to run asynchronously if they are expected to take more than two seconds to complete.

> The `return_timeout` query parameter is available with asynchronous API calls and can convert an asynchronous call to complete synchronously. Refer to Asynchronous processing using the Job object for more information.

Security

The security provided with the REST API is based primarily on the existing security features available with ONTAP. The following security is used by the API:

Transport Layer Security

All traffic sent over the network between the client and ONTAP LIF is typically encrypted using TLS, based on the ONTAP configuration settings.

Client authentication

The same authentication options available with ONTAP System Manager and the Network Manageability SDK can also be used with the ONTAP REST API.

HTTP authentication

At an HTTP level, for example when accessing the ONTAP REST API directly, there are two authentication options as described below. In each case, you need to create an HTTP authorization header and include it with each request.

Option	Description
HTTP basic authentication	The ONTAP username and password are concatenated with a colon. The string is converted to base64 and included in the request header.
OAuth 2.0	Beginning with ONTAP 9.14, you can request an access token from an external authorization server and include it as a bearer token in the request header.

For more details about OAuth 2.0 and how it is implemented in ONTAP, see Overview of the ONTAP OAuth 2.0 implementation. Also see Prepare to use the workflows below at this site.

ONTAP authorization

ONTAP implements a role-based authorization model. The account you use when accessing the ONTAP REST API or API documentation page should have the proper authority.

Input variables controlling an API request

You can control how an API call is processed through parameters and variables set in the

HTTP request.

HTTP methods

The HTTP methods supported by the ONTAP REST API are shown in the following table.

> Not all the HTTP methods are available at each of the REST endpoints. Also, both PATCH and DELETE can be used on a collection. See *Object references and access* for more information.

HTTP method	Description
GET	Retrieves object properties on a resource instance or collection.
POST	Creates a new resource instance based on the supplied input.
PATCH	Updates an existing resource instance based on the supplied input.
DELETE	Deletes an existing resource instance.
HEAD	Effectively issues a GET request but only returns the HTTP headers.
OPTIONS	Determine what HTTP methods are supported at a specific endpoint.

Path variables

The endpoint path used with each REST API call can include various identifiers. Each ID corresponds to a specific resource instance. Examples include cluster ID and SVM ID.

Request headers

You must include several headers in the HTTP request.

Content-type

If the request body includes JSON, this header must be set to `application/json`.

Accept

This header should be set to `application/hal+json`. If it is instead set to `application/json` none of the HAL links will be returned except a link needed to retrieve the next batch of records. If the header is anything else aside from these two values, the default value of the `content-type` header in the response will be `application/hal+json`.

Authorization

Basic authentication must be set with the user name and password encoded as a base64 string. For example:

```
Authorization: Basic YWRtaW46cGV0ZXJzb24=.
```

Request body

The content of the request body varies depending on the specific call. The HTTP request body consists of one of the following:

- JSON object with input variables

- Empty JSON object

Filtering objects

When issuing an API call with the GET method, you can limit or filter the returned objects based on any attribute using a query parameter.

Parsing and interpreting query parameters

A set of one or more parameters can be appended to the URL string beginning after the `?` character. If more than one parameter is provided, the query parameters are split based on the `&` character. Each key and value in the parameter are split at the = character.

For example, you can specify an exact value to match using the equal sign:

```
<field>=<value>
```

For a more complex query, the additional operator is placed after the equal sign. For example, to select the set of objects based on a specific field that is greater than or equal to some value, the query would be:

```
<field>=>=<value>
```

Filtering operators

In addition to the examples provided above, additional operators are available to return objects over a range of values. A summary of the filtering operators supported by the ONTAP REST API is shown in the table below.

> Any fields that are not set are generally excluded from matching queries.

Operator	Description
=	Equal to
<	Less than
>	Greater than
<=	Less than or equal to
>=	Greater than or equal to
!	Not equal to
*	Greedy wildcard

You can also return a collection of objects based on whether a specific field is set or not set by using the `null` keyword or its negation `!null` as part of the query.

Workflow examples

Some examples are included below from the REST API workflows at this site.

- List disks

  Filter based on the `state` variable to select the spare disks.

Requesting specific object fields

By default, issuing an API call using GET returns only the attributes that uniquely identify the object or objects, along with a HAL self link. This minimum set of fields acts as a key for each object and varies based on the object type. You can select additional object properties using the `fields` query parameter in the following ways:

- Common or standard fields

  Specify `fields=*` to retrieve the most commonly used object fields. These fields are typically maintained in local server memory or require little processing to access. These are the same properties returned for an object after using GET with a URL path key (UUID).

- All fields

  Specify `fields=**` to retrieve all the object fields, including those requiring additional server processing to access.

- Custom field selection

  Use `fields=<field_name>` to specify the exact field you want. When requesting multiple fields, the values must be separated using commas without spaces.

> As a best practice, you should always identify the specific fields you want. You should only retrieve the set of common fields or all fields when needed. Which fields are classified as common, and returned using `fields=*`, is determined by NetApp based on internal performance analysis. The classification of a field might change in future releases.

Sorting objects in the output set

The records in a resource collection are returned in the default order defined by the object. You can change the order using the `order_by` query parameter with the field name and sort direction as follows:

```
order_by=<field name> asc|desc
```

For example, you can sort the type field in descending order followed by id in ascending order:

```
order_by=type desc, id asc
```

Note the following:

- If you specify a sort field but don’t provide a direction, the values are sorted in ascending order.
- When including multiple parameters, you must separate the fields with a comma.

Pagination when retrieving objects in a collection

When issuing an API call using GET to access a collection of objects of the same type, ONTAP attempts to return as many objects as possible based on two constraints. You can control each of these constraints using additional query parameters on the request. The first constraint reached for a specific GET request terminates the request and therefore limits the number of records returned.

If a request ends before iterating over all the objects, the response contains the link needed to retrieve the next batch of records.

Limiting the number of objects

By default, ONTAP returns a maximum of 10,000 objects for a GET request. You can change this limit using the `max_records` query parameter. For example:

```
max_records=20
```

The number of objects actually returned can be less than the maximum in effect, based on the related time constraint as well as the total number of objects in the system.

Limiting the time used to retrieve the objects

By default, ONTAP returns as many objects as possible within the time allowed for the GET request. The default timeout is 15 seconds. You can change this limit using the `return_timeout` query parameter. For example:

```
return_timeout=5
```

The number of objects actually returned can be less than the maximum in effect, based on the related constraint on the number of objects as well as the total number of objects in the system.

Narrowing the result set

If needed, you can combine these two parameters with additional query parameters to narrow the result set. For example, the following returns up to 10 ems events generated after the specified time:

```
time=> 2018-04-04T15:41:29.140265Z&max_records=10
```

You can issue multiple requests to page through the objects. Each subsequent API call should use a new time value based on the latest event in the last result set.

Size properties

The input values used with some API calls as well as certain query parameters are numeric. Rather than provide an integer in bytes, you can optionally use a suffix as shown in the following table.

Suffix	Description
KB	KB Kilobytes (1024 bytes) or kibibytes
MB	MB Megabytes (KB x 1024 bytes) or mebibytes
GB	GB Gigabytes (MB x 1024 bytes) or gibibytes
TB	TB Terabytes (GB x 1024 bytes) or tebibytes
PB	PB Petabytes (TB x 1024 bytes) or pebibytes

Related information

- Object references and access

Interpreting an API response

Each API request generates a response back to the client. You should examine the response to determine whether it was successful and retrieve additional data as needed.

HTTP status code

The HTTP status codes used by the ONTAP REST API are described below.

Code	Reason phrase	Description
200	OK	Indicates success for calls that do not create a new object.
201	Created	An object is successfully created. The location header in the response includes the unique identifier for the object.
202	Accepted	A background job has been started to perform the request, but has not completed yet.
400	Bad request	The request input is not recognized or is inappropriate.
401	Unauthorized	User authentication has failed.
403	Forbidden	Access is denied due to an authorization error.
404	Not found	The resource referred to in the request does not exist.
405	Method not allowed	The HTTP method in the request is not supported for the resource.
409	Conflict	An attempt to create an object failed because a different object must be created first or the requested object already exists.
500	Internal error	A general internal error occurred at the server.

Response headers

Several headers are included in the HTTP response generated by the ONTAP.

Location

When an object is created, the location header includes the complete URL to the new object including the unique identifier assigned to the object.

Content-type

This will normally be `application/hal+json`.

Response body

The content of the response body resulting from an API request differs based on the object, processing type, and the success or failure of the request. The response is always rendered in JSON.

- Single object

  A single object can be returned with a set of fields based on the request. For example, you can use GET to retrieve selected properties of a cluster using the unique identifier.

- Multiple objects

  Multiple objects from a resource collection can be returned. In all cases, there is a consistent format used, with `num_records` indicating the number of records and records containing an array of the object instances. For example, you can retrieve the nodes defined in a specific cluster.

- Job object

  If an API call is processed asynchronously, a Job object is returned which anchors the background task. For example, the PATCH request used to update the cluster configuration is processed asynchronously and returns a Job object.

- Error object

  If an error occurs, an Error object is always returned. For example, you will receive an error when attempting to change a field not defined for a cluster.

- Empty JSON object

  In certain cases, no data is returned and the response body includes an empty JSON object.

HAL linking

The ONTAP REST API uses HAL as the mechanism to support Hypermedia as the Engine of Application State (HATEOAS). When an object or attribute is returned that identifies a specific resource, a HAL-encoded link is also included allowing you to easily locate and determine additional details about the resource.

Errors

If an error occurs, an error object is returned in the response body.

Format

An error object has the following format:

```
"error": {
"message": "<string>",
"code": <integer>[,
"target": "<string>"]
}
```

You can use the code value to determine the general error type or category, and the message to determine the specific error. When available, the target field includes the specific user input associated with the error.

Common error codes

The common error codes are described in the following table. Specific API calls can include additional error codes.

Code		Description
1	409	An object with the same identifier already exists.

Code		Description
2	400	The value for a field has an invalid value or is missing, or an extra field was provided.
3	400	The operation is not supported.
4	405	An object with the specified identifier cannot be found.
6	403	Permission to perform the request is denied.
8	409	The resource is in use.

Asynchronous processing using the Job object

After issuing an API request that is designed to run asynchronously, a job object is always created and returned to the caller. The job describes and anchors a background task that processes the request. Depending on the HTTP status code, you must retrieve the state of the job to determine if the request was successful.

Refer to API reference to determine which API calls are designed to be performed asynchronously.

Controlling how a request is processed

You can use the `return_timeout` query parameter to control how an asynchronous API call is processed. There are two possible outcomes when using this parameter.

Timer expires before the request completes

For valid requests, ONTAP returns a 202 HTTP status code along with the job object. You must retrieve the state of the job to determine if the request completed successfully.

Request is completed before the timer expires

If the request is valid and completes successfully before the time expires, ONTAP returns a 200 HTTP status code along with the job object. Because the request is completed synchronously, as indicated by the 200, you do not need to retrieve the job state.

> The default value for the `return_timeout` parameter is zero seconds. Therefore, if you don't include the parameter, the 202 HTTP status code is always returned for a valid request.

Querying the Job object associated with an API request

The Job object returned in the HTTP response contains several properties. You can query the state property in a subsequent API call to determine if the request completed successfully. A Job object is always in one of the following states:

Non-terminal states

- Queued
- Running

- Paused

Terminal states

- Success
- Failure

General procedure for issuing an asynchronous request

You can use the following high-level procedure to complete an asynchronous API call. This example assumes the `return_timeout` parameter is not used, or that the time expires before the background job completes.

1. Issue an API call that is designed to be performed asynchronously.
2. Receive an HTTP response 202 indicating acceptance of a valid request.
3. Extract the identifier for the Job object from the response body.
4. Within a timed loop, perform the following in each cycle:
   a. Get the current state of the Job.
   b. If the Job is in a non-terminal state, perform loop again.
5. Stop when the Job reaches a terminal state (success, failure).

Related information

- Update cluster contact
- Get job instance

Object references and access

The resource instances or objects exposed through the ONTAP REST API can be referenced and accessed in several different ways.

Object access paths

At a high level, there are two path types when accessing an object:

- Primary

  The object is the primary or direct target of the API call.

- Foreign

  The object is not the primary reference of the API call, but rather is linked to from the primary object. It is therefore a foreign or downstream object and referenced through a field in the primary object.

Accessing an object using the UUID

Every object is assigned a unique identifier when it is created, which in most cases is a 128-bit UUID. The assigned UUID values are immutable and are used internally within ONTAP to access and manage the resources. Because of this, the UUID generally provides the fastest and most stable way to access objects.

For many of the resource types, a UUID value can be provided as part of the path key in the URL to access a

specific object. For example, you can use the following to access a node instance: `/cluster/nodes/{uuid}

Accessing an object using an object property

In addition to a UUID, you can also access an object using an object property. In most cases, it is convenient to use the name property. For example, you can use the following query parameter in the URL string to access a node instance by its name: `/cluster/nodes?name=node_one`. In addition to a query parameter, a foreign object can be accessed through a property in the primary object.

While you can use the name or other property to access an object instead of the UUID, there are several possible disadvantages:

- The name field is not immutable and can be changed. If the name of an object is changed before accessing an object, the wrong object will be returned or an object access error will fail.

  This issue can occur with a POST or PATCH method on a foreign object or with a GET method on a primary object.

- ONTAP must translate the name field into the corresponding UUID. This is a type of indirect access which can become a performance issue.

In particular, a performance degradation is possible when one or more of the following is true:

- GET method is used
- A large collection of objects is accessed
- A complex or elaborate query is used

Cluster versus SVM context

There are several REST endpoints that support both a cluster and SVM. When using one of these endpoints, you can indicate the context of the API call through the `scope=[svm|cluster]` value. Examples of endpoints supporting a dual context include IP interfaces and security roles.

The scope value has a default value base on the properties provided for each API call.

Using PATCH and DELETE on a collection of objects

Every REST endpoint supporting PATCH or DELETE on a resource instance also supports the same method on a collection of objects. The only requirement is that at least one field must be provided through a query parameter in the URL string. When issuing a PATCH or DELETE over a collection, this is equivalent to doing the following internally:

- Query-based GET to retrieve the collection
- Serial sequence of PATCH or DELETE calls on each object in the collection

The time out for the operation can be set by `return_timeout` with a default of 15 seconds. If not completed before the timeout, the response includes a link to the next object. You must reissue the same HTTP method using the next link to continue the operation.

Performance metrics for storage resources

ONTAP collects performance metrics about selected SVM storage objects and protocols, and reports this information through the REST API. You can use this data to monitor the performance of an ONTAP system.

For a given storage object or protocol, the performance data falls into three categories:

- IOPS
- Latency
- Throughput

Within each category, one or more of the following types of data is available:

- Read ®
- Write (W)
- Other (O)
- Total (T)

The following table summarizes the performance data available through the ONTAP REST API, including the release when it was added. Refer to the REST API online documentation page at your ONTAP system for more information.

Storage object or protocol	IOPS	Latency	Throughput	ONTAP release
Ethernet port	Not applicable	Not applicable	RWT	9.8
FC port	RWOT	RWOT	RWT	9.8
IP interface	Not applicable	Not applicable	RWT	9.8
FC interface	RWOT	RWOT	RWT	9.8
NVMe namespace	RWOT	RWOT	RWOT	9.8
Qtree statistics	Raw RWOT	Not applicable	Raw RWOT	9.8
Volume Flexcache	RWOT	RWOT	RWT	9.8
Node – process utilization	Process utilization as a numerical value	Process utilization as a numerical value	Process utilization as a numerical value	9.8
Cloud volume	RWOT	RWOT	Not appliable	9.7
LUN	RWOT	RWOT	RWOT	9.7
Aggregate	RWOT	RWOT	RWOT	9.7
SVM NFS protocol	RWOT	RWOT	RWT	9.7
SVM CIFS protocol	RWOT	RWOT	RWT	9.7
SVM FCP protocol	RWOT	RWOT	RWT	9.7
SVM iSCSI protocol	RWOT	RWOT	RWT	9.7

Storage object or protocol	IOPS	Latency	Throughput	ONTAP release
SVM NVMe protocol	RWOT	RWOT	RWT	9.7
Cluster	RWOT	RWOT	RWOT	9.6
Volumes	RWOT	RWOT	RWOT	9.6

Copyright information

Copyright © 2024 NetApp, Inc. All Rights Reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means—graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system—without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

LIMITED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (b)(3) of the Rights in Technical Data -Noncommercial Items at DFARS 252.227-7013 (FEB 2014) and FAR 52.227-19 (DEC 2007).

Data contained herein pertains to a commercial product and/or commercial service (as defined in FAR 2.101) and is proprietary to NetApp, Inc. All NetApp technical data and computer software provided under this Agreement is commercial in nature and developed solely at private expense. The U.S. Government has a non-exclusive, non-transferrable, nonsublicensable, worldwide, limited irrevocable license to use the Data only in connection with and in support of the U.S. Government contract under which the Data was delivered. Except as provided herein, the Data may not be used, disclosed, reproduced, modified, performed, or displayed without the prior written approval of NetApp, Inc. United States Government license rights for the Department of Defense are limited to those rights identified in DFARS clause 252.227-7015(b) (FEB 2014).

Trademark information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.